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RESEARCH ARTICLE

Smoother Integral Sliding Mode Filter for RSSI-Based Localization

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ABSTRACT This study advances our previous research on radio frequency (RF) source localization using unmanned aerial vehicles (UAVs) through a novel geometrical Received Signal Strength Indicator (RSSI) approach. Our methodology utilizes four UAVs, split into two groups, which adjust their positions until the power received by each UAV matches that of its counterpart, allowing the calculation of intersecting lines that pinpoint the source location. Recognizing the challenges posed by environmental noise which can alter signal values leading to inaccurate estimations, we introduce a smoother integral sliding mode filter. This enhanced filter incorporates the integral of the error into the sliding mode control framework to address the high-frequency chattering typically associated with sliding mode filters in noisy environments. This adaptation not only reduces the chattering effect but also stabilizes the UAV's response to fluctuating signal strengths, resulting in more accurate and reliable localization. Simulation results demonstrate a marked improvement in localization accuracy and robustness against noise.

INDEX TERMS Localization, received signal strength indicator, sliding mode filter, unmanned aerial vehicles.

I. INTRODUCTION

This The advent of Unmanned Aerial Vehicles (UAVs) has revolutionized a myriad of sectors, ranging from agriculture and disaster management to surveillance and logistics. These versatile platforms have become indispensable tools in the Internet of Things (IoT) ecosystem, offering innovative solutions to traditional challenges [1]. Among the myriad of applications, localization and tracking stand out as critical functionalities, especially in contexts where precision and reliability are paramount, such as air traffic control, remote sensing, and intelligence, surveillance, and reconnaissance (ISR) operations [2].

Traditional ground-based localization methods have been gradually overshadowed by the capabilities offered by UAVs. The mobility, flexibility, and elevated vantage points of UAVs enhance signal reception and reduce propagation uncertainties caused by terrestrial obstacles, thereby improving the accuracy of RF target identification [3]. However, the dynamic nature of UAV operations introduces new challenges, particularly in signal processing for localization in environments where GPS signals are weak or non-existent [4].

Our previous research introduced a new geometrical localization method leveraging the Received Signal Strength Indicator (RSSI) for UAV groups [5]. This method, distinguished by its simplicity and efficiency, does not require knowledge of the path loss exponent or transmit power, making it particularly suitable for RF target localization with minimal hardware requirements. Despite its promising results, the initial approach was based on an idealistic assumption of signal transmission without reflection or distortion, a condition rarely met in real-world scenarios.

Real-world environments are inherently noisy, with signal distortion and reflection being commonplace. These factors significantly impact the accuracy of RSSI-based localization methods, necessitating the development of approaches that can effectively handle such conditions. Recent studies have attempted to address these challenges by incorporating advanced signal processing techniques and machine learning algorithms to enhance the robustness of RSSI measurements. Methods in [5] and [6] have shown promise in improving

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accuracy, but they often come with increased computational complexity and the need for extensive training data, making them less practical for real-time UAV applications.

Recognizing this gap, our current study aims to refine our previous localization scheme [7] by incorporating environmental noise into the signal processing framework. This involves the adoption of a sliding mode filter, a robust control method known for its effectiveness in dealing with uncertainties and disturbances, including noise [8], [9].

The sliding mode filter approach addresses the key challenge of signal noise by filtering the noisy RSSI data [10], [11], thereby enhancing the reliability of the localization process. However, the application of this method in a UAV context revealed a high-frequency chattering phenomenon in the calculated angle values, posing a significant challenge for the practical implementation of UAV movements [12]. To overcome this, we introduced the integral of the error in the angle calculation process, resulting in smoother and more feasible desired angle values for UAV navigation.

In the pursuit of advancing UAV-based localization techniques, particularly in environments fraught with noise and uncertainties, Sliding Mode Filtering (SMF) emerges as a compelling solution [13]. Originating from the principles of Sliding Mode Control (SMC), a robust control strategy distinguished by its resilience against disturbances and model in-accuracies, SMF applies these principles to the domain of state estimation and filtering [14], [15], [16]

Sliding Mode Filtering capitalizes on the construction of a sliding surface, typically a function of the estimation error, and employs a discontinuous control law to drive the es-timation errors towards this surface in finite time [8]. This approach ensures that, once on the sliding surface, the system's estimated states converge to their true values despite the presence of noise or other external perturbations. The inherent robustness of SMF against uncertainties makes it particularly suitable for RSSI-based localization tasks where the signal can be significantly affected by multipath effects, atmospheric conditions, and other noise sources.

Recent advancements in sliding mode theory have further enhanced the efficacy and applicability of SMF in complex systems. For instance, in [17] authors discuss state-of-the-art developments in sliding mode techniques, including filters that have been tailored for systems with non-linear dynamics and uncertain parameters—characteristics common in UAV localization scenarios. Furthermore, the adaptability of sliding mode filters in real-time applications underscores their potential for improving the accuracy and reliability of UAV-based RF source localization [9].

In RSSI-based systems, precision and the ability to adjust to signal fluctuations are crucial for maintaining accurate localization and tracking. The refined filter mechanism enhances overall system performance by ensuring smoother transition dynamics, making it highly effective in diverse engineering applications where robust and adaptable control systems are crucial for optimal functionality.

Incorporating SMF into UAV-based localization systems promises significant improvements in the robustness and precision of localization under uncertain conditions. By effectively mitigating the impact of noise on RSSI measurements, sliding mode filters pave the way for more accurate and dependable localization solutions, addressing one of the primary challenges faced in GPS-denied environments [18].

This paper expands on our previous work by detailing the methodology, simulation results, and comparative analysis of the enhanced localization scheme. By addressing the critical issue of environmental noise, this study contributes to the field of UAV-based localization, offering a more accurate, robust, and practical solution for real-world applications.

The remainder of the paper is arranged as follows: the background of RSSI and sliding mode filter are covered in Section II. Section III provides the methodology of the suggested approach; Section IV discusses simulation results to validate the suggested method; and Section V offers the study's conclusion.

II. BACKGROUND AND PRELIMINARIES

To address the challenges posed by noisy RSSI measurements in UAV-based RF source localization, this study introduces a background that incorporates sliding mode filters into the localization process. The sliding mode filter, known for its robustness against disturbances and uncertainties, is applied to the RSSI data to produce more accurate distance estimations, thereby improving the localization accuracy of the RF source.

RSSI is a method that determines the power of signal strength at the receiver based on a variety of parameters, including transmitting power, receiver and transmitter antenna characteristics, and distance between the two. The distance between the transmitter and the receiver can be calculated using the measured signal strength. In space, the power of the signal varies with the seperation between the transmitter and the receiver. The follow-ing Friis equation [19] can be used to measure this:

$$P_r = P_t D_r D_t \left(\frac{\lambda}{4\pi d}\right)^{\gamma} + N(0, \sigma) \tag{1}$$

In this formula, P_t represents the power of the signal transmitted, d is the separation between the transmitter and receiver, γ denotes the path loss exponent, and λ is the wavelength of the signal transmitted. To account for errors from reflections, measurements, and other factors, an additive Gaussian noise N is incorporated.

This paper introduces a geometrical approach based on RSSI for localizing RF devices when GPS is unavailable. The development of this method addresses two significant obstacles: the lack of prior location data for the target and reliance solely on signal strength measurements by UAVs, without additional signal information or direction-finding capabilities. To tackle these issues, it is presumed that a direct visual connection always exists between the UAVs and

the target, and that the UAVs can exchange data flawlessly, without any communication constraints.

Sliding Mode Filter (SMF) is an advanced estimation technique that applies the principles of sliding mode control to the problem of filtering and state estimation in dynamic systems. At its core, SMF aims to robustly estimate the states of a system by driving the estimation errors towards a predefined sliding surface and ensuring they remain on this surface despite disturbances or model uncertainties [14].

The performance of the sliding mode filter is influenced by several critical parameters, including the design of the sliding surface, the gain matrix, the coefficient of any integral terms, and the characteristics of system noise. The sliding surface determines the robustness and convergence speed of the system, while the gain matrix controls the response to deviations from this surface, where improper tuning can lead to chattering or instability [14]. Noise characteristics, such as amplitude and frequency, also play a significant role in the filter's effectiveness, as does the system's sampling rate, which impacts the filter's ability to respond to changes and suppress noise. The sensitivity of the filter to these parameters requires careful tuning to optimize performance, particularly in dynamic environments where real-time adaptation may be necessary. Recent advancements in sliding mode control, including adaptive techniques, offer promising approaches to dynamically adjust these parameters for enhanced performance in varying conditions [15].

Consider a system described by the given differential equation:

$$\dot{x} = f(x,t) + B(x,t)u + w \tag{2}$$

where $x \in \mathbb{R}^n$ represents the state vector of the system. $f(x,t)$ is a known function representing the system dynamics. $B(x,t)$ is the input matrix and is the control input. w represents the process noise or disturbances affecting the system.

A. SLING SURFACE DESIGN

The sliding surface $S(x)$ is chosen such that the system's behaviour is satisfactory when the system's trajectory lies on this surface. It is typically a function of the estimation error $e = x - \hat{x}$, where $\hat{x}$ is the estimated state. The design of $S(x)$ aims to ensure that the error dynamics are stable and converge to zero, i.e., $S(x) = 0$. A common choice for $S(x)$ in sliding mode filters is

$$S(x) = C(e) = C(x - \hat{x}) \tag{3}$$

where C is a matrix or function designed to shape the error dynamics.

The control law in sliding mode is designated to be discontinuous, forcing the system's states onto the sliding surface in finite time and maintaining it there despite disturbances. The control law involves a term that becomes active when the system's states deviate from the sliding surface:

$$u = u_0 - K.sign(S(x)) \tag{4}$$

where L is a gain matrix designed to incorporate the measurement feedback (y) into estimation process. y represents the measured output, which is used to update the estimate $\hat{x}$

III. METHODOLOGY

The methodology is detailed in subsequent subsections. Initially, the precise 3D Cartesian positions for each UAV are determined to enable their movement along two orthogonal circles—one in the x-y plane and another in the x-z plane, each at a radius R from the center. Then, the angles of rotation necessary for the UAVs to orbit the circles while preserving a separation of 2R between them are computed. These rotational angles are derived based on the variances in the power of the signals received. These angles are then used to calculate 3 orthogonal vectors for target localization.

While UAVs offer significant advantages in a wide range of applications, their onboard processors typically lack the capacity for handling computationally intensive tasks, and efficient energy use is crucial for maintaining their mobility and operational endurance. To address these limitations, our method offloads the computational burden by having UAVs transmit the received signal strength data to a more powerful central computer. This central system then calculates the optimal positions for the UAVs by analysing the differences in signal strength within the groups, iteratively updating and transmitting new position commands to the UAVs until they achieve equal signal strength. This approach not only conserves the UAVs' processing power and energy but also ensures precise and synchronized adjustments to their positions, enhancing the overall accuracy and reliability of the localization process which is summarized in Figure 1.

A. DERIVATION OF FOUR UAVs' DESIRED INITIAL POSTURE

Imagine four UAVs, each outfitted with RSSI receivers, starting from random positions. The initial target locations for these UAVs are determined as follows:

$$\begin{aligned}
x_1^d(t_0) &= x_c - R\cos(\alpha(t_0)) \\
y_1^d(t_0) &= y_c - R\sin(\alpha(t_0)) \quad z_1^d = z_c \\
x_3^d(t_0) &= x_c + R\cos(\alpha(t_0)) \\
y_3^d(t_0) &= y_c + R\sin(\alpha(t_0)) \quad z_3^d = z_c \\
x_2^d(t_0) &= x_c - R\sin(\beta(t_0)) \\
y_2^d &= y_c \quad z_3^d(t_0) = z_c - R\cos(\beta(t_0)) \\
x_4^d(t_0) &= x_c + R\cos(\beta(t_0)) \\
y_4^d &= y_c \quad z_4^d(t_0) = z_c + R\sin(\beta(t_0))
\end{aligned} \tag{5}$$

where $C(x_c, y_c, z_c)$ is the center of the formation, R is the radius of the circles, $\alpha(t_0)$, and $\beta(t_0)$ are the initial rotation angles given in Figure 2 defined by user and t_0 is the initial time. D1 and D3 UAVs form a group in x-y plane while the D2 and D4 UAVs form another group in x-z plane.

The initial positions (5) of the UAVs, operating along the x-y and x-z axes, are determined based on two key conditions. Firstly, each UAV must be located at a distance R

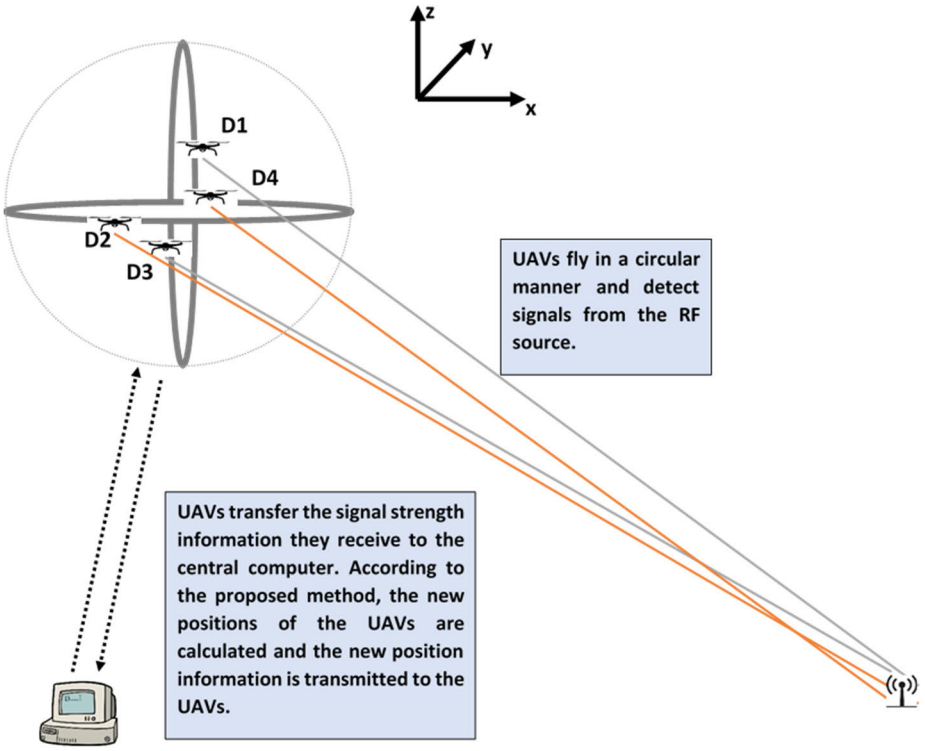


FIGURE 1. The UAVs detect RF signal and communicate with central unit for new position estimates.

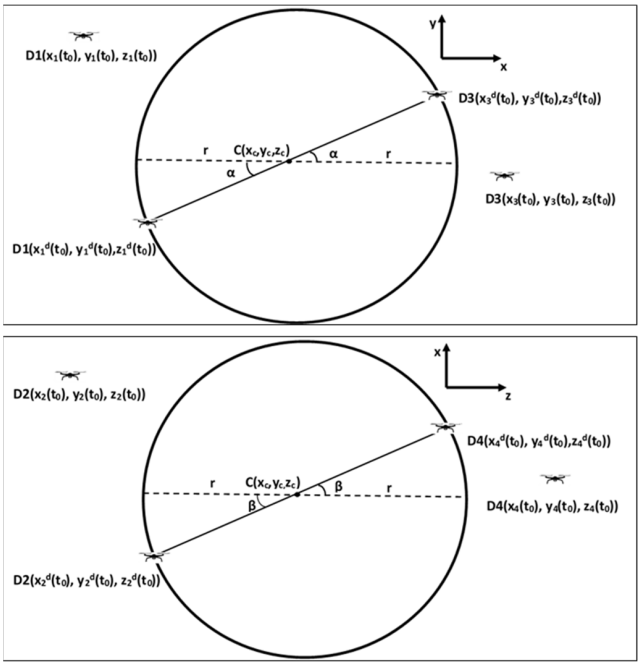


FIGURE 2. The UAVs' starting and intended positions [5].

from the center. Secondly, the separation between any two UAVs should be $2R$. Consequently, the UAVs navigate along a predefined circle, maintaining a constant distance from each other as given in Figure 2. Details on how the rotational angles are calculated are provided in the following subsection.

In this study, for simplicity, we assume the presence of a single UAV on each axis. However, scaling up the number of UAVs, for instance to 4n, could potentially improve performance by providing a more robust average of the signals received from each group. Nevertheless, such an increase in the UAV fleet would come with considerable trade-offs. The additional UAVs would not only drive up the financial costs, but would also amplify the communication and computational demands, posing significant challenges in managing the coordination and data processing for a larger fleet. These factors must be carefully considered when balancing performance enhancements against the practical implications of scalability.

B. ROTATING ANGLE ESTIMATION WITH SMOOTHER INTEGRAL SLIDING MODE FILTER

The power differences are specified in the following way to determine the target's orientation in both the vertical and horizontal planes.

$$\begin{aligned} e_1 &= P_3 - P_1 \\ e_2 &= P_4 - P_2 \end{aligned} \tag{6}$$

where $P_1, \ldots, P_4$ are acquired powers by $D_1, \ldots, D_4$ respectively. e_1 is obtained from the difference in signal strengths of the first and third UAVs, e_2 and is obtained from the power differences of the second and fourth UAVs. The error dynamics are then presented as

$$\dot{e}_1 = \dot{P}_3 - \dot{P}_1$$

$$\dot{e}_2 = \dot{P}_4 - \dot{P}_2 \tag{7}$$

and

$$\begin{aligned} s_1 &= \dot{e}_1 + \lambda e_1 \\ s_2 &= \dot{e}_2 + \lambda e_2 \end{aligned} \tag{8}$$

s_1 and s_2 are the sliding surfaces which defines the condition or manifold that the system's trajectory is forced to reach and maintain. In this study, we want each UAV in the group to receive equal power, that is, the power difference between them to go to zero.

when $s_1 \to 0, \mathrm{e}_1 \to 0$
if $s_1 = 0,\ \dot{e}_1 + \lambda e_1 = 0,\quad \dot{e}_1 = -\lambda e_1$ and $e_1(\mathrm{t}) = e^{-\lambda(\mathrm{t})}$
if $\lambda \gg 0, \mathrm{e}_1 \to 0$
and similar approach to
if $s_2 = 0, \dot{e}_2 + \lambda e_2 = 0,\ \dot{e}_2 = -\lambda e_2$ and $e_2(\mathrm{t}) = e^{-\lambda(\mathrm{t})}$
if $\lambda \gg 0, \mathrm{e}_2 \to 0$

To determine the target's horizontal direction, two UAVs traveling along the x-y axis are utilized, and to determine the target's vertical direction, UAVs traveling along the x-z plane are employed. In order to achieve a zero difference in the signal strengths received by the two UAVs, the derivatives of α and β are computed. Sliding mode filter is employed and derivative of α was defined as the function of s_1 and derivative of β as the function of s_2.

The desired rotating angle dynamics are defined as

$$\begin{aligned} \dot{\alpha} &= k_{s_1} sign(s_1) \\ \dot{\beta} &= k_{s_1} sign(s_2) \end{aligned} \tag{9}$$

with $k_{s_1} > 0$ and $k_{s_2} > 0$ are design parameters.

Chattering in sliding mode control (and by extension, sliding mode filtering) occurs due to the inherent discontinuous nature of the control law applied to ensure the system's states reach and stay on the predefined sliding surface. This discontinuity arises from the use of a sign function or a similar discontinuous function in the control law, which leads to rapid switching of the control input.

In order to mitigate this effect, smoother integral sliding mode filter is developed in this study. The error integral is incorporated into the algorithm of the traditional sliding mode filter to substantially mitigate chattering. This key modification smoothens the control actions, which not only reduces noise but also enhances the system's durability. By integrating the error, the filter achieves improved stability and robustness, enabling consistent performance even in the face of system uncertainties and external disturbances. The integral component of the error also significantly improves tracking accuracy and promotes faster error convergence, which is particularly valuable in applications such as target tracking with RSSI.

Then the angle dynamics are calculated as

$$\dot{\alpha} = k_{s_1} sign(\mathrm{s}_1) + \int_{t_0}^{t} e_1(\tau)\mathrm{d}\tau$$

$$\dot{\beta} = k_{s_1} sign(\mathrm{s}_2) + \int_{t_0}^{t} e_2(\tau)\mathrm{d}\tau \tag{10}$$

Then the current rotating angles are found as

$$\begin{aligned} \alpha(t) &= \alpha(t_0) + \int_{t_0}^{t} \dot{\alpha}(\tau)\mathrm{d}\tau \\ \beta(t) &= \beta(t_0) + \int_{t_0}^{t} \dot{\beta}(\tau)\mathrm{d}\tau \end{aligned} \tag{11}$$

Changing α and β in time (11) will automatically change the desired locations of the UAVs according to (12). This change continues until, e_1 and e_2 converges to zero or a predefined and tolerable error bound.

$$\begin{aligned} x_1^d(t) &= x_c - R\cos(\alpha(t)) \quad y_1^d(t) = y_c - R\sin(\alpha(t)) \\ z_1^d &= z_c \\ x_3^d(t) &= x_c + R\cos(\alpha(t)) \quad y_3^d(t) = y_c + R\sin(\alpha(t)) \\ z_3^d &= z_c \\ x_2^d(t) &= x_c - R\sin(\beta(t)) \quad y_2^d = y_c \\ z_3^d(t) &= z_c - R\cos(\beta(t)) \\ x_4^d(t) &= x_c + R\cos(\beta(t)) \quad y_4^d = y_c \\ z_4^d(t) &= z_c + R\sin(\beta(t)) \end{aligned} \tag{12}$$

After the angle values are determined, the vector perpendicular to the center of the sphere and the vectors formed by both groups of objects will be determined. This vector will pass through the source position. To calculate the location of the target, it is necessary to find at least two vectors. Different vectors can also be calculated by moving the center of the system to a different location. The intersection point of these vectors calculated at different positions will give the source position. The calculation of these vectors is given in the next subsection.

C. DETERMINING THE LINES PASS THROUGH SOURCE AND SPHERE CENTER

Let us assume that the sphere center was changed N times and N different angle values were calculated according to the final position of the UAVs. The final angles for i^{th} iteration, where $i = 1, \ldots, N$, at their final value are represented as $\alpha\left(t_{f\alpha_i}\right)$ and $\beta\left(t_{f\beta_i}\right)$ where $t_{f\alpha_i}$ and $t_{f\beta_i}$ are time instants that the angles reached their final value. The final positions of the UAVs in each iteration are defined as

$$\begin{aligned} x_1^d\left(t_{f\alpha_i}\right) &= x_c - R\cos\left(\alpha\left(t_{f\alpha_i}\right)\right) \\ y_1^d\left(t_{f\alpha_i}\right) &= y_c - R\sin\left(\alpha\left(t_{f\alpha_i}\right)\right) \quad z_1^d = z_c \\ x_3^d\left(t_{f\alpha_i}\right) &= x_c + R\cos\left(\alpha\left(t_{f\alpha_i}\right)\right) \\ y_3^d\left(t_{f\alpha_i}\right) &= y_c + R\sin\left(\alpha\left(t_{f\alpha_i}\right)\right) \quad z_3^d = z_c \\ x_2^d\left(t_{f\beta_i}\right) &= x_c - R\sin\left(\beta\left(t_{f\beta_i}\right)\right) \\ y_2^d &= y_c \quad z_3^d\left(t_{f\beta_i}\right) = z_c - R\cos\left(\beta\left(t_{f\beta_i}\right)\right) \end{aligned}$$

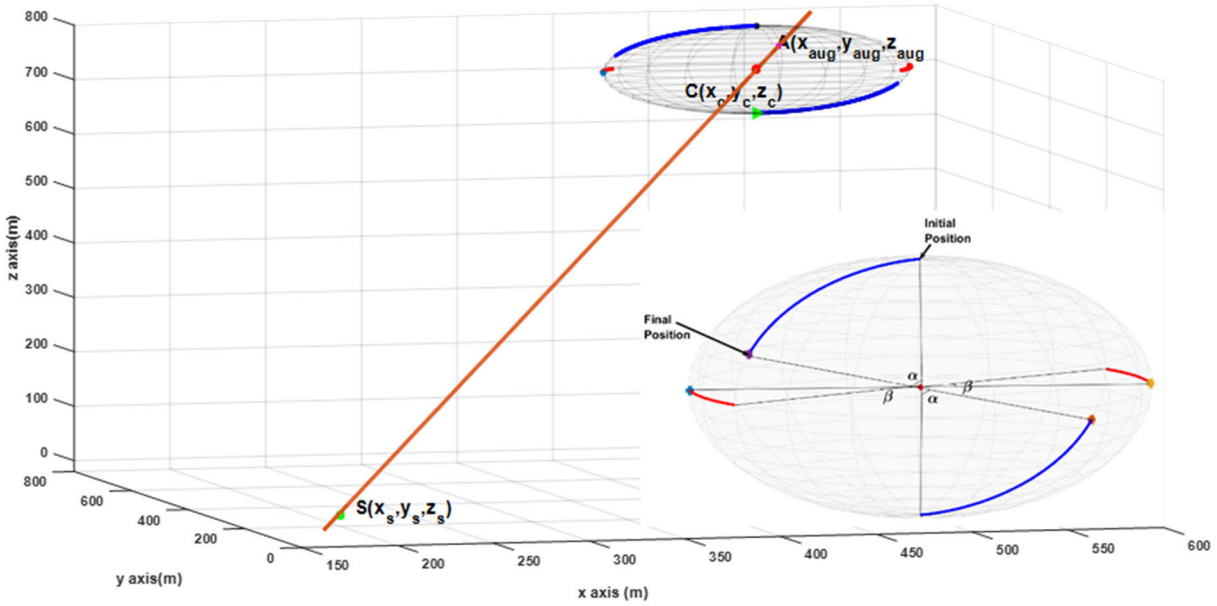


FIGURE 3. Rotation of UAVs and line passing through source location and formation center [5].

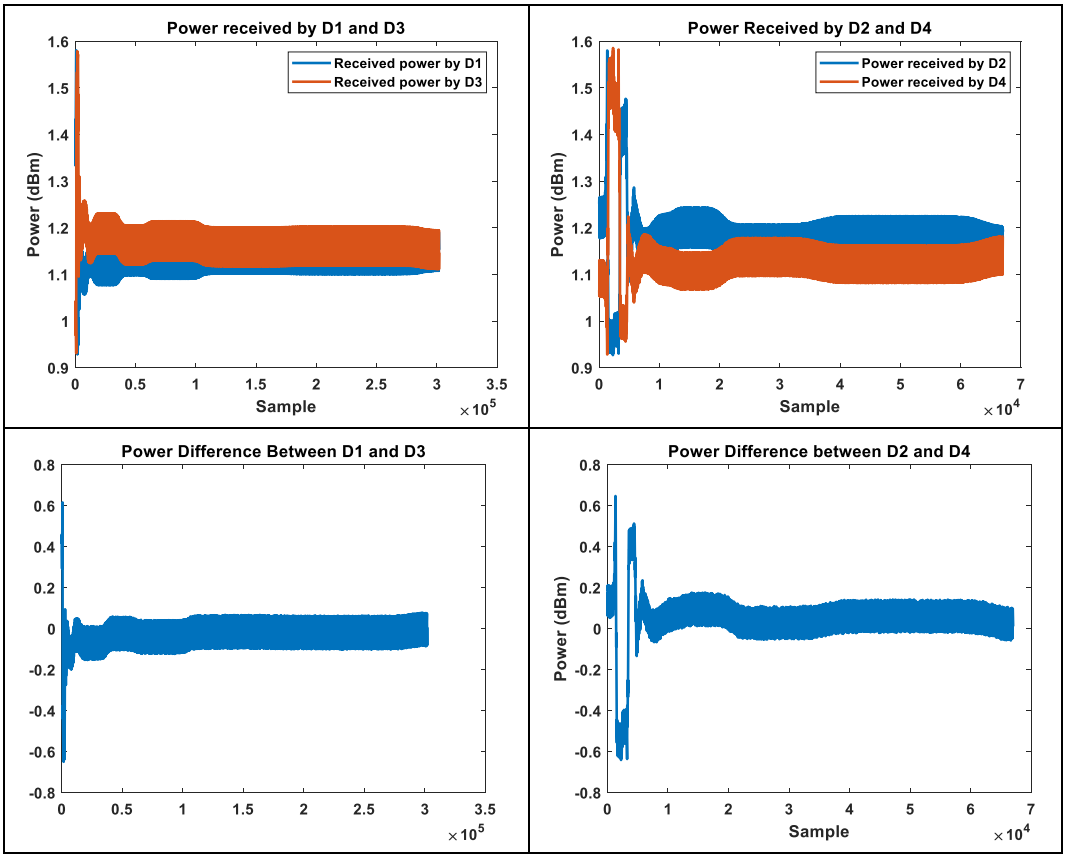


FIGURE 4. Received signal strengths of UAVs until they reach equal received power.

$$x_4^d\left(t_{f\beta_i}\right) = x_c + R\cos\left(\beta\left(t_{f\beta_i}\right)\right)$$
$$y_4^d = y_c \quad z_4^d\left(t_{f\beta_i}\right) = z_c + R\sin\left(\beta\left(t_{f\beta_i}\right)\right) \quad (13)$$

To calculate the source position, first define a vector through D1 and D3, similarly with D2 and D4. These vectors are

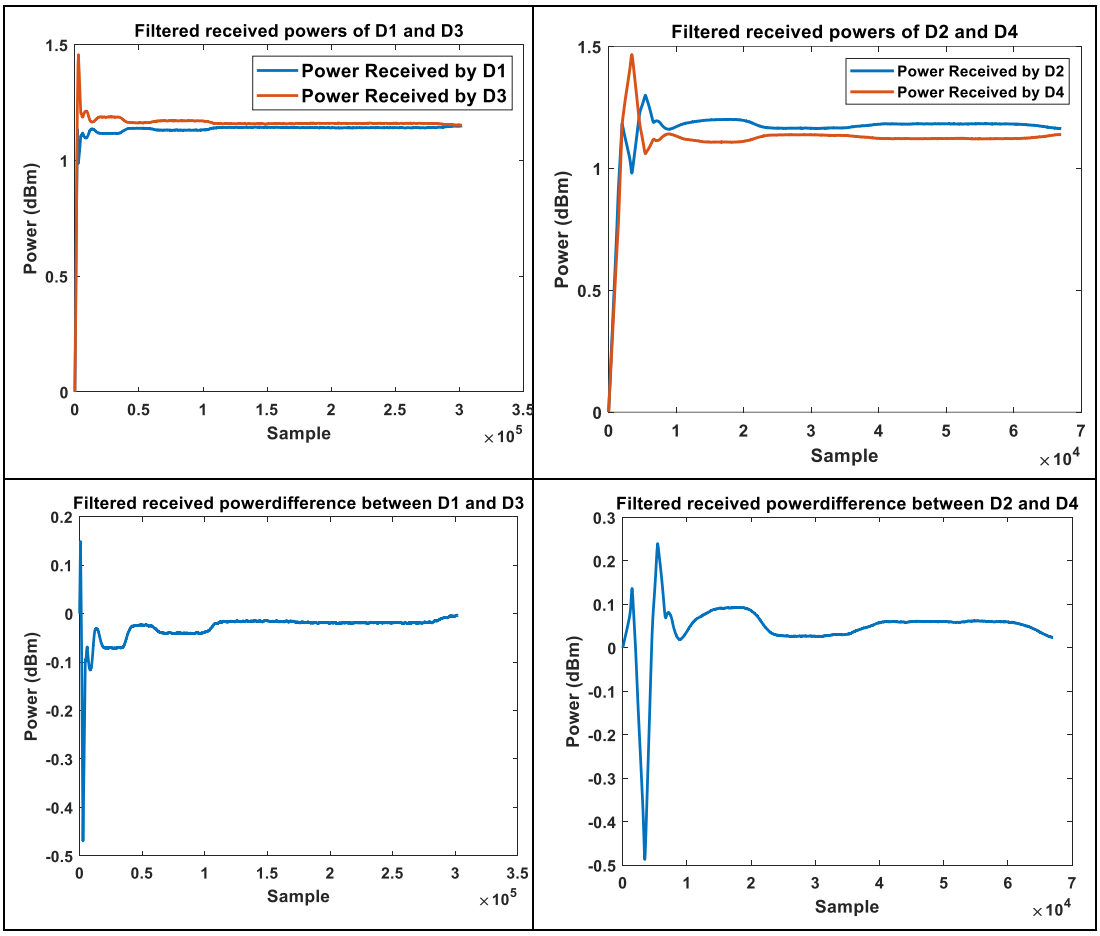


FIGURE 5. Filtered received powers of the UAVs.

perpendicular to each other and they intersect in the center of the sapphire that the UAV's move around. The vector perpendicular to these two vectors will pass through target position. The vector defined between D1 and D3 is

$$\overrightarrow{v_1} = [R\cos\left(\alpha\left(t_{f\alpha_i}\right)\right),\ R\sin\left(\alpha\left(t_{f\alpha_i}\right)\right),\ 0] \tag{14}$$

and similarly, between D2 and D4 is

$$\overrightarrow{v_2} = [R\cos\left(\beta\left(t_{f\beta_i}\right)\right), 0, R\sin\left(\beta\left(t_{f\beta_i}\right)\right)] \tag{15}$$

The cross product $\overrightarrow{v_3} = \overrightarrow{v_1} \times \overrightarrow{v_2}$ is calculated as follows.

$$\overrightarrow{v_3} = \begin{bmatrix} R^2\sin\left(\alpha\left(t_{f\alpha_i}\right)\right)\sin\left(\beta\left(t_{f\beta_i}\right)\right), \\ -R^2\cos\left(\alpha\left(t_{f\alpha_i}\right)\right)\sin\left(\beta\left(t_{f\beta_i}\right)\right), \\ -R^2\sin\left(\alpha\left(t_{f\alpha_i}\right)\right)\cos\left(\beta\left(t_{f\beta_i}\right)\right) \end{bmatrix} \tag{16}$$

Given $\overrightarrow{v_3}$ and knowing the line passes through the target position is

$$\begin{aligned} x &= R^2\sin\left(\alpha\left(t_{f\alpha_i}\right)\right)\sin\left(\beta\left(t_{f\beta_i}\right)\right)t \\ y &= -R^2\cos\left(\alpha\left(t_{f\alpha_i}\right)\right)\sin\left(\beta\left(t_{f\beta_i}\right)\right)t \\ z &= z - R^2\sin\left(\alpha\left(t_{f\alpha_i}\right)\right)\cos\left(\beta\left(t_{f\beta_i}\right)\right)t \end{aligned} \tag{17}$$

where t is a parameter. The given procedure is demonstrated in Figure 3.

IV. RESULTS

The suggested method's performance is described in this section. For simulation experiments, target localization within a 1×1 km^2 search area was taken into consideration. It is presumed that the target is lying on the ground. UAVs first look through the GPS-denied area until they come cross an RF signal. Following that, they revolve around the designated circles until each group of UAVs has an equal amount of power.

After UAVs detect the presence of the target, they continue their rotation movements until they receive equal signal strength. However, as can be seen from the figure, the fluctuation in signal strengths is a bit high. This causes erroneous results in location determination which is demonstrated in Figure 4.

Figure 5 shows the signal strengths with the smoother integral sliding mode filter applied, demonstrating significantly reduced fluctuations compared to the unfiltered. data. This stabilization is critical as it directly correlates with more precise location estimates of the RF source. By filtering out the noise and minimizing the influence of environmental variables, the smoother integral sliding mode filter ensures that the UAVs achieve a more consistent and reliable signal

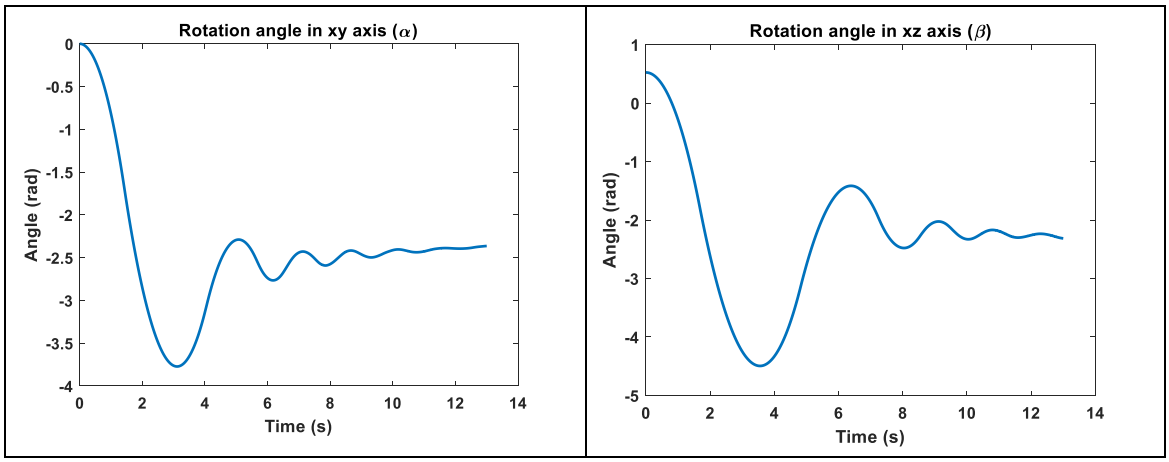


FIGURE 6. Rotation angles of the UAVs in their predetermined plane.

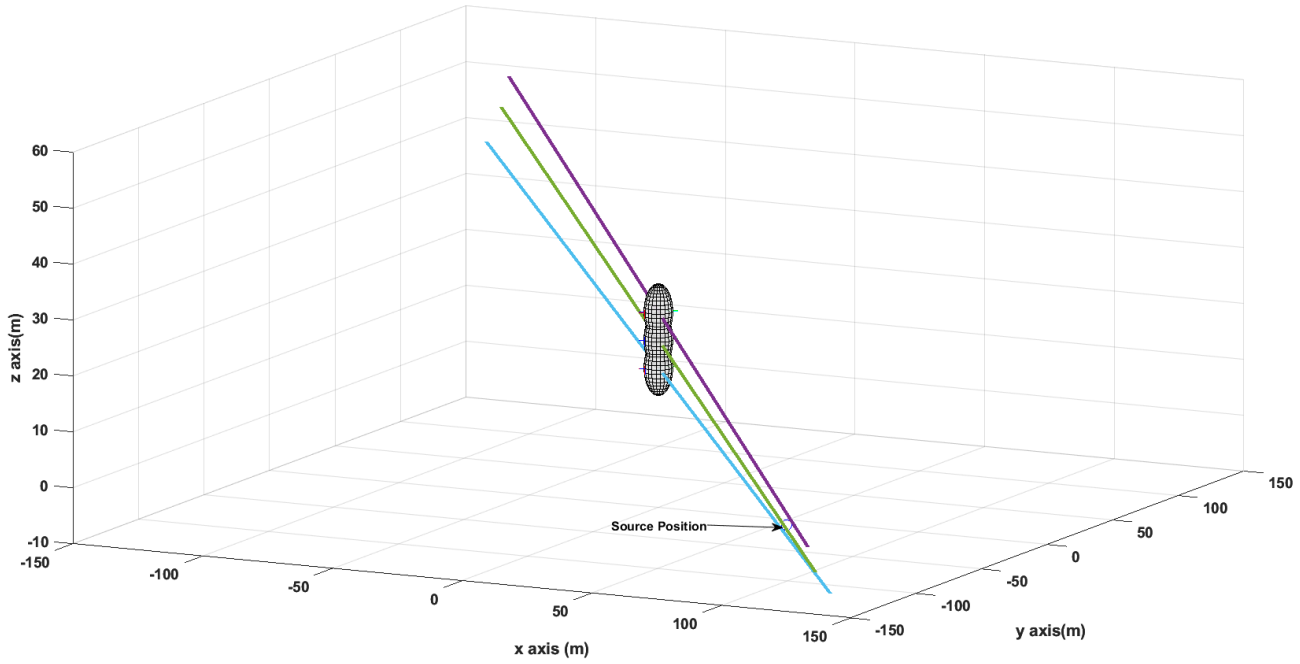


FIGURE 7. Source localization using proposed algorithm.

reception. The fixing of the rotation angles at the end of the algorithm, when equal signal strength is reached, is shown in Figure 6.

When the UAVs in the groups reach equal signal strength, the line section passing through the center of the sphere and the target location is calculated as specified in (13). The rotation movement of the UAVs to receive equal power and the calculated line are shown Figure 2. As stated in previous work [5], a single accurate target location is insufficient to calculate it. For this, the system center is shifted, the UAVs come to the position where they will receive equal power, and the line calculation is performed again. The point where these lines intersect, provides the source location.

Figure 7 illustrates the process of target detection when the received signal strength deviates from the expected theoretical signal power, as detailed in equation (1). The analysis was conducted with the sphere center positioned at three distinct locations. By calculating the nearest common point among the perpendicular lines generated from each scanning position [5], this point was identified as the estimated location of the target. A series of 100 Monte Carlo simulations were conducted, with the RF source's position randomly varied in each instance. The results showed an average error of 23.7 meters, attributed to a noise variance of 4% in signal power.

V. CONCLUSION

This paper has presented an advanced method for RF source localization using UAVs, employing a smoother integral sliding mode filter to enhance the accuracy and reliability of

RSSI-based localization techniques. Through the integration of the error integral into the sliding mode control framework, we have successfully mitigated the common issues of chattering and noise sensitivity, which are prevalent in conventional sliding mode filters.

Our simulations demonstrate a significant improvement in localization accuracy and robustness against environmental noise, underscoring the effectiveness of the smoother integral sliding mode filter. This approach not only stabilizes the signal fluctuations but also ensures more consistent UAV

positioning, which is crucial for accurate source detection and localization in complex, real-world environments.

This study primarily focuses on the application of the smoother integral sliding mode filter under Gaussian noise conditions, commonly encountered in UAV operations. However, future work will expand on these tests by incorporating different noise structures and additional environmental factors, such as multipath interference and atmospheric conditions. These extended studies aim to further enhance the robustness of our localization method under real-world scenarios.

To further enhance the filter's performance in highly dynamic environments, future work can incorporate adaptive or learning-based approaches. These methods will enable the filter to dynamically adjust its parameters in real-time, ensuring robust performance even as signal characteristics rapidly change.

While the proposed method demonstrates significant potential, there are certain considerations to keep in mind. The requirement for a minimum of four UAVs, may present challenges in scenarios where resources are constrained. Additionally, the method assumes reliable communication between UAVs, which could face occasional disruptions due to environmental factors such as signal interference or obstacles. Although the smoother integral sliding mode filter effectively reduces the impact of environmental noise, the method's performance might still be influenced by extreme noise conditions or highly complex environments, such as dense urban areas. These limitations may create new study topics in the future.

This study contributes to the field of UAV-based localization by providing a robust, practical solution that can be deployed in GPS-denied environments, potentially benefiting a wide range of applications from military operations to emergency response.

REFERENCES

[1] H. Shakhatreh, A. H. Sawalmeh, A. Al-Fuqaha, Z. Dou, E. Almaita, I. Khalil, N. S. Othman, A. Khreishah, and M. Guizani, "Unmanned aerial vehicles (UAVs): A survey on civil applications and key research challenges," *IEEE Access*, vol. 7, pp. 48572–48634, 2019.

[2] A. Guerra, B. Guerreiro, and J. Sousa, "UAV navigation for target tracking and 3D localization," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 54, no. 4, pp. 1884–1899, Jan. 2018.

[3] Y. Zeng, R. Zhang, and T. J. Lim, "Wireless communications with unmanned aerial vehicles: Opportunities and challenges," *IEEE Commun. Mag.*, vol. 54, no. 5, pp. 36–42, May 2016.

[4] R. Kapoor, K. Bhamidipati, and S. A. Smolka, "Localization challenges in UAV communication networks," *IEEE Commun. Mag.*, vol. 57, no. 1, pp. 70–75, May 2019.

[5] A. Dey, T. Chattopadhyay, and T. Paul, "Adaptive RSSI-based indoor localization using machine learning," *IEEE Sensors J.*, vol. 21, no. 6, pp. 7981–7988, May 2021.

[6] H. Liu, X. Zhang, and L. Zhou, "RSSI-based indoor localization using unmanned aerial vehicle in wireless sensor networks," *Sensors*, vol. 17, no. 10, pp. 1–22, Jun. 2017.

[7] N. Güzey, "RF source localization using multiple UAVs through a novel geometrical RSSI approach," *Drones*, vol. 6, no. 12, p. 417, Dec. 2022.

[8] Y. Shtessel, C. Edwards, L. Fridman, and A. Levant, *Sliding Mode Control and Observation*. Cham, Switzerland: Springer, 2014.

[9] V. Utkin and A. Poznyak, *Advances in Sliding Mode Control: Concept, Theory and Implementation*. Cham, Switzerland: Springer, 2013.

[10] S. Jin, R. Kikuuwe, and M. Yamamoto, "Real-time quadratic sliding mode filter for removing noise," *Adv. Robot.*, vol. 26, nos. 8–9, pp. 877–896, May 2012.

[11] M. T. S. Aung, Z. Shi, and R. Kikuuwe, "A new noise-reduction filter with sliding mode and low-pass filtering," in *Proc. IEEE Conf. Control Appl. (CCA)*, Oct. 2014, pp. 1029–1034.

[12] F. Demim, A. Rouigueb, H. Belaidi, A. Messaoui, K. Bensseghieur, A. Allam, M. Benatia, A. Nouri, and A. Nemra, "Smooth sliding mode control based technique of an autonomous underwater vehicle based localization using obstacle avoidance strategy," in *Proc. 20th Int. Conf. Informat. Control, Autom. Robot.*, 2023, pp. 529–537.

[13] S. Modali and S. Ghosh, "Sliding mode-based guidance for UAV landing on a stationary or moving ground vehicle," *IFAC-PapersOnLine*, vol. 53, no. 1, pp. 453–458, 2020.

[14] V. Utkin, J. Guldner, and J. Shi, *Sliding Mode Control in Electro-Mechanical Systems*. Boca Raton, FL, USA: CRC Press, 2009.

[15] S. Spurgeon, "Adaptive sliding mode control for nonlinear systems: Real-time parameter estimation and control," *IEEE Control Syst. Mag.*, vol. 28, no. 5, pp. 19–34, Apr. 2018.

[16] C. Edwards and S. K. Spurgeon, *Sliding Mode Control: Theory and Applications*. New York, NY, USA: Taylor & Francis, 1998.

[17] L. Fridman, J. Moreno, and R. Iriarte, *Sliding Modes After the First Decade of the 21st Century: State of the Art*. Cham, Switzerland: Springer, 2011.

[18] V. Utkin, *Sliding Modes in Control and Optimization*. Cham, Switzerland: Springer, 2013.

[19] J. H. Lee and R. M. Buehrer, "Fundamentals of received signal strength-based position location," in *Handbook of Position Location: Theory, Practice, and Advance*. Hoboken, NJ, USA: Wiley, 2012, pp. 359–394.



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